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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2017 Trial Examination

FORM VI

MATHEMATICS EXTENSION 1

Friday 4th August 2017

General Instructions

- Reading time — 5 minutes
- Writing time — 2 hours
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 70 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Reference sheet
- Candidature — 125 boys

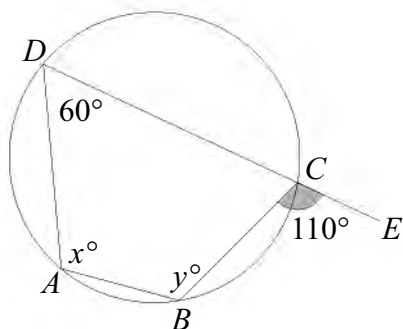
Examiner

FMW

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE



Suppose $ABCD$ is a cyclic quadrilateral with DC produced to E . What are the values of x and y ?

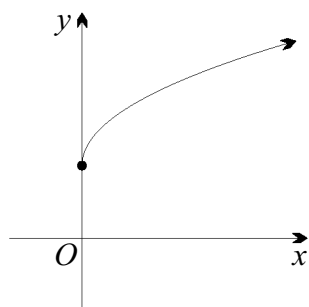
- (A) $x = 120, y = 110$
- (B) $x = 110, y = 110$
- (C) $x = 120, y = 120$
- (D) $x = 110, y = 120$

QUESTION TWO

Let $A = (-3, 2)$ and $B = (4, -7)$. The interval AB is divided externally in the ratio $5 : 3$ by the point $P(x, y)$. What is the value of x ?

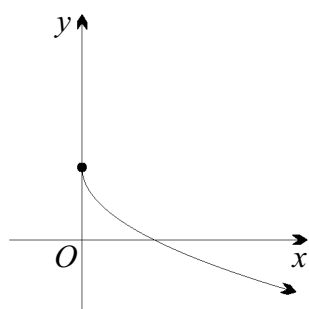
- (A) $14\frac{1}{2}$
- (B) 13
- (C) $1\frac{3}{8}$
- (D) $-13\frac{1}{2}$

QUESTION THREE

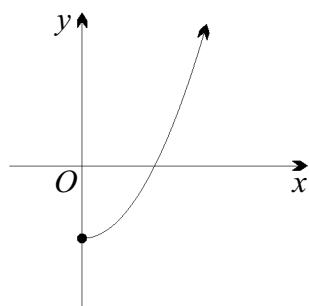


The diagram shows the graph of $y = f(x)$. Which diagram shows the graph of $y = f^{-1}(x)$?

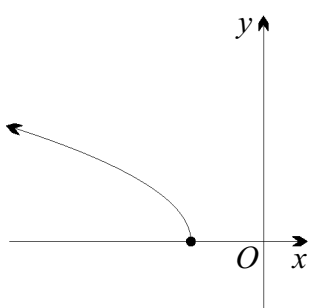
(A)



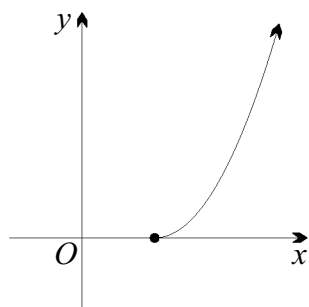
(B)



(C)



(D)



QUESTION FOUR

What is the derivative of $\sin^{-1} 3x$?

(A) $\frac{1}{3\sqrt{1-9x^2}}$

(B) $\frac{-1}{3\sqrt{1-3x^2}}$

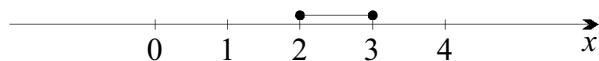
(C) $\frac{3}{\sqrt{1-9x^2}}$

(D) $\frac{3}{\sqrt{1-3x^2}}$

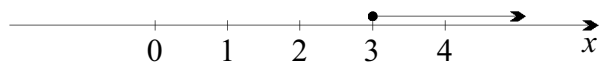
QUESTION FIVE

Which number line graph shows the correct solution to $\frac{x}{x-2} \geq 3$?

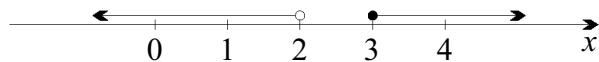
(A)



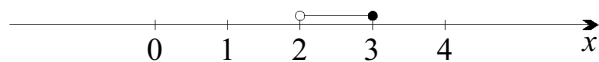
(B)



(C)



(D)



QUESTION SIX

What is the domain of the function $y = 4 \sin^{-1} \frac{x}{3}$?

- (A) $-3 \leq x \leq 3$
- (B) $-\frac{1}{3} \leq x \leq \frac{1}{3}$
- (C) $-2\pi \leq x \leq 2\pi$
- (D) $-\frac{\pi}{8} \leq x \leq \frac{\pi}{8}$

QUESTION SEVEN

What is the maximum value of $P = 6 \cos \theta + 4 \sin \theta$?

- (A) 10
- (B) 6
- (C) $2\sqrt{13}$
- (D) $2\sqrt{5}$

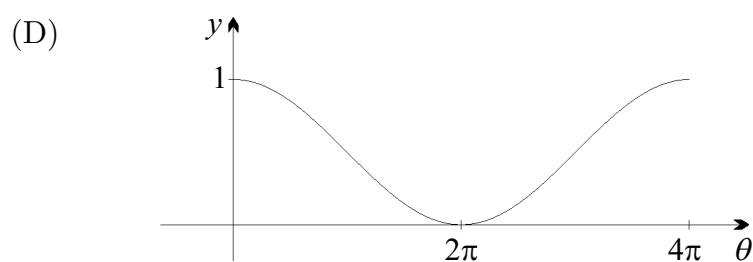
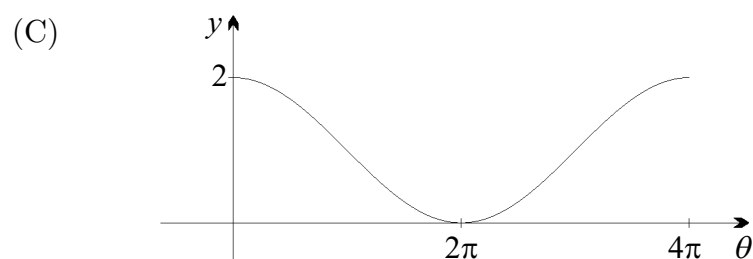
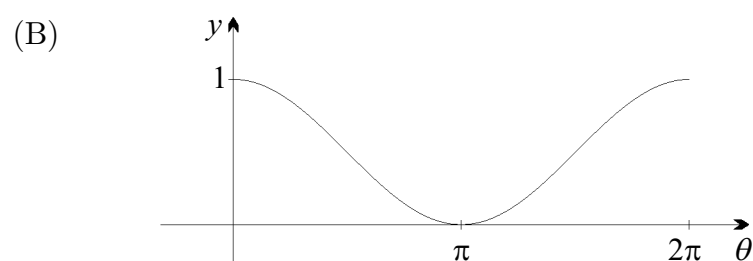
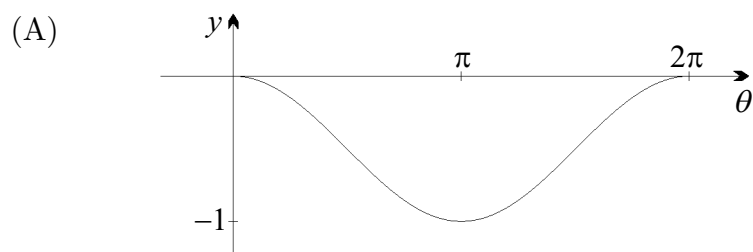
QUESTION EIGHT

A particle moves on a line so that its distance from the origin at time t seconds is x cm and its acceleration is given by $\frac{d^2x}{dt^2} = 10 - 2x^3$. If v represents the velocity of the particle, and the particle changes direction 1 cm on the negative side of the origin, which of the following equations is correct?

- (A) $v^2 = 20x - x^4$
- (B) $v^2 = 20x - x^4 + 21$
- (C) $v = 10x - \frac{1}{2}x^4$
- (D) $v = 10x - \frac{1}{2}x^4 + 11\frac{1}{2}$

QUESTION NINE

Which of the diagrams below best represents the graph of $y = \cos^2 \frac{1}{2}\theta$?



QUESTION TEN

What is the coefficient of z^3 in the expansion of $(1 + z + z^2)^5$?

- (A) 10
- (B) 20
- (C) 30
- (D) 40

_____ End of Section I _____

Examination continues next page ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.

Marks

(a) Find the exact value of $\sin \frac{\pi}{8} \cos \frac{\pi}{8}$. 2

(b) Evaluate $\sin^{-1}(\sin \frac{4\pi}{3})$. 1

(c) Show that $\lim_{x \rightarrow 0} \frac{\tan \frac{x}{2}}{x} = \frac{1}{2}$. 1

(d) Find the following integrals:

(i) $\int \frac{4x}{16 + x^2} dx$ 1

(ii) $\int \frac{3}{9 + x^2} dx$ 1

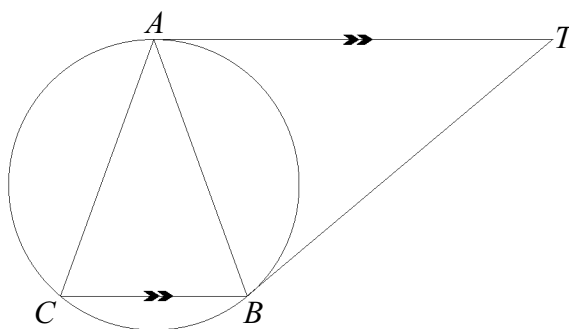
(iii) $\int \frac{-1}{\sqrt{25 + x}} dx$ 1

(e) Write down a general solution of the equation $\sin x = -\frac{1}{2}$. 1

(f) If a , b and c are the roots of the equation $3x^3 + 4x^2 - 5x - 8 = 0$, find the value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$. 2

(g) By expanding, find the greatest coefficient in the expansion of $(4x + 3)^4$. 2

(h) 3



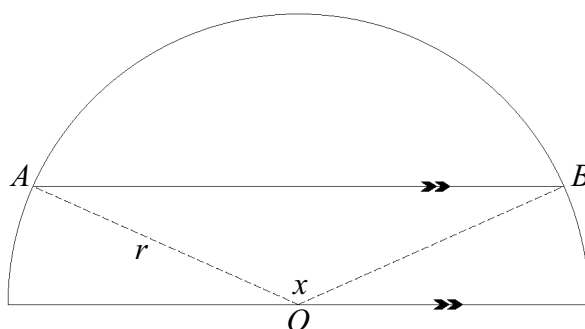
Tangents touching a circle at A and B respectively, intersect at T . Point C is on the circle and $AT \parallel CB$. Prove that $AB = AC$.

Examination continues overleaf ...

QUESTION TWELVE (15 marks) Use a separate writing booklet.**Marks**

- (a) An object is put in a freezer to cool. After t minutes, its temperature is $T^{\circ}\text{C}$. The freezer is at a constant temperature of -8°C . The object's temperature T decreases according to the differential equation $\frac{dT}{dt} = -k(T + 8)$, where k is a positive constant.
- (i) Show that $T = Ae^{-kt} - 8$, where A is a constant, is a solution of the differential equation. 1
- (ii) If the object cools from an initial temperature of 40°C to 30°C in half an hour, find the values of A and k . 2
- (iii) When will the temperature of the object be 0°C ? Give your answer correct to the nearest hour. 1
- (iv) Explain what will happen to T eventually. 1

(b)



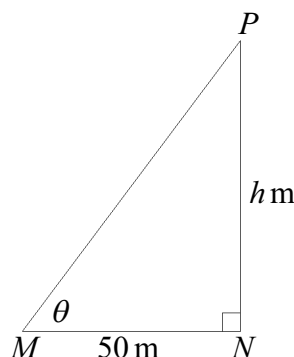
The diagram above shows a semi-circle of radius r with centre O . Chord AB is drawn parallel to the base such that it divides the semi-circle into two parts of equal area. Chord AB subtends an angle of x radians at the centre O .

- (i) Show that $\sin x = x - \frac{\pi}{2}$. 1
- (ii) The equation has a root near $x = 2$. Use one application of Newton's method to find a better approximation for this root, writing your answer correct to three significant figures. 2

(c) (i) Use the substitution $u = 3x + 1$ to show that $\int_0^1 \frac{x}{(3x+1)^2} dx = \frac{2}{9} \ln 2 - \frac{1}{12}$. 2

- (ii) Hence find the volume of the solid formed when the region bounded by the curve $y = \frac{6\sqrt{x}}{3x+1}$, the x -axis and the line $x = 1$ is rotated about the x -axis. Give your answer in exact form. 1

(d)



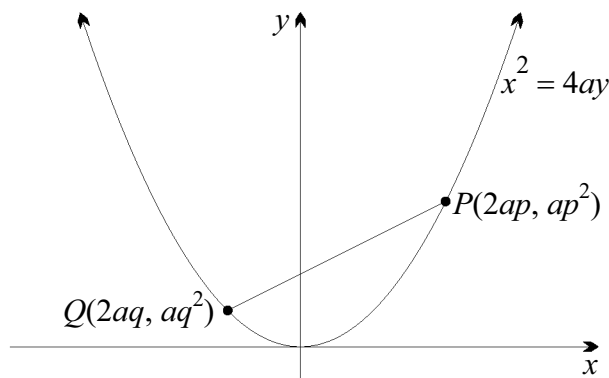
Bowie jumps out of a helicopter and by the time he reaches the position P , h metres above the ground, he is falling at a constant rate of 150 kilometres per hour. Point N is on the ground directly below P and M lies 50 metres from N . The angle of elevation of P from M is θ radians.

(i) Show that $\frac{dh}{d\theta} = \frac{50}{\cos^2 \theta}$. 1

- (ii) Find the rate of decrease of the angle of elevation when Bowie reaches a height of 1200 metres. Give your answer in radians per second. 3

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.**Marks**

(a)

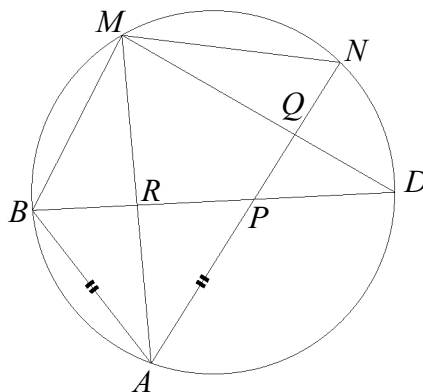


The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola with equation $x^2 = 4ay$.

- (i) Find the coordinates of M , the midpoint of PQ . 1
 - (ii) Show that the equation of the chord PQ is $y = \frac{1}{2}(p + q)x - apq$. 1
 - (iii) If the chord always passes through the point $(0, 2a)$, find the equation of the locus of M . 2
- (b) A particle moves along a straight line and its displacement, x centimetres, from a fixed point O at a given time t seconds is given by $x = 2 + \cos^2 t$.
- (i) Show that its acceleration is given by $\ddot{x} = 10 - 4x$. 2
 - (ii) Explain why the motion is simple harmonic. 1
 - (iii) Find the centre, amplitude and period of the motion. 2
- (c) The polynomial $P(x)$ is given by $P(x) = x^3 - mx^2 + mx - 1$, where m is a constant.
- (i) Show that $(x - 1)$ is a factor of $P(x)$. 1
 - (ii) Hence find a quadratic factor of $P(x)$. 2
 - (iii) Hence find the set of values of m for which all the roots of the equation $P(x) = 0$ are real. 2
 - (iv) If $m = 3$, the graph of $y = P(x)$ is a transformation of the graph of $y = x^3$. Describe this transformation. 1

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks****3**

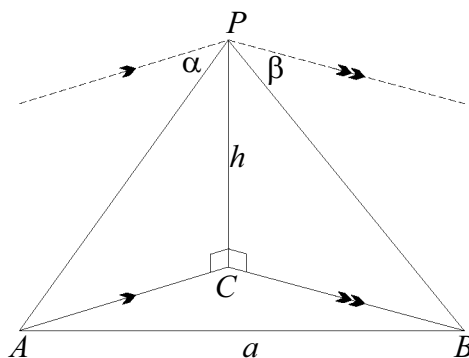
(a)



The diagram above shows a cyclic quadrilateral $ABMN$. Point P lies on AN such that $AB = AP$ and BP produced meets the circle again at D and AM at R . The chord MD intersects AN at Q .

Copy the diagram and show that $QPRM$ is a cyclic quadrilateral.

(b)



The diagram above shows two points A and B on level ground. B is a metres due east of A . A tower, of height h metres, is also on the same level ground and its bearing is $N\theta E$ and $N\phi W$ from A and B respectively. From the top of the tower P , the angle of depression of A is α and of B is β .

(i) Prove that $h \sin(\theta + \phi) = a \cos \phi \tan \alpha$.

2

(ii) Prove that $h^2(\cot^2 \alpha - \cot^2 \beta) - 2ha \cot \alpha \sin \theta + a^2 = 0$.

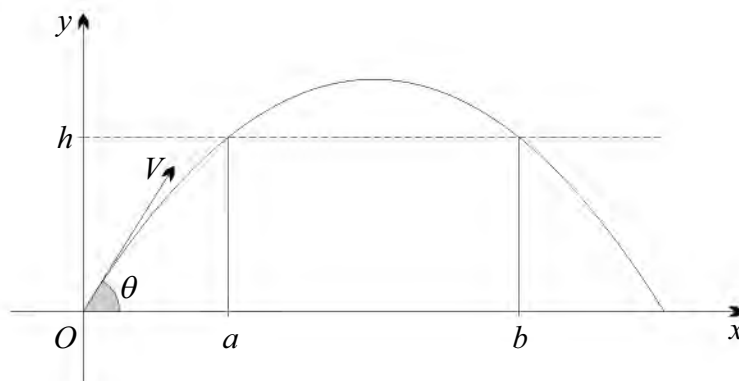
2

(c) If $f^{(n)}(x)$ denotes the n th derivative of $f(x) = \frac{1}{x}$, prove by mathematical induction

3

that $f^{(n)}(x) = \frac{(-1)^n n!}{x^{n+1}}$ for all positive integers n .

(d)



A particle is fired from O with initial velocity V m/s at an angle θ to the horizontal. The particle just clears two thin vertical towers of height h metres at horizontal distances of a metres and b metres from O .

The equations of motion of the particle are $x = Vt \cos \theta$ and $y = Vt \sin \theta - \frac{1}{2}gt^2$.
(Do NOT prove these equations.)

(i) Show that $V^2 = \frac{a^2 g(1 + \tan^2 \theta)}{2(a \tan \theta - h)}$. **2**

(ii) Hence show that $\tan \theta = \frac{h(a+b)}{ab}$. **2**

(iii) Hence show that $\tan \theta = \tan \alpha + \tan \beta$, where α and β are the angles of elevation from O to the tops of the towers. **1**

————— End of Section II —————

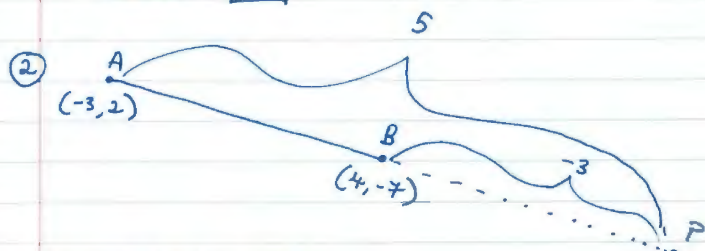
END OF EXAMINATION

Extension I TRIAL 2017 SOLUTIONS

Multiple Choice:

- ① $x = 110$ (exterior angle of cyclic quadrilateral ABCD)
 $y = 120$ (opposite angles of cyclic quadrilateral ABCD)

choose **D**

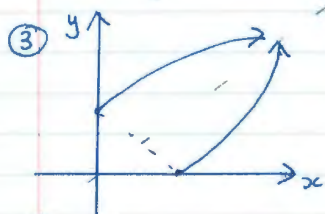


$$x = \frac{(-3)(-3) + 5(4)}{5 - 3}$$

$$= \frac{29}{2}$$

$$= 14\frac{1}{2}$$

choose **A**



swap x/y
 reflect across line $y = x$

choose **D**

④ $y = \sin^{-1}(3x)$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(3x)^2}} \times 3$$

$$= \frac{3}{\sqrt{1-9x^2}}$$

choose **C**

⑤ $(x-2)^2 \times \frac{x}{x-2} > 3(x-2)^2$

$$x(x-2) > 3(x-2)^2$$

$$3(x-2)^2 - x(x-2) \leq 0$$

$$(x-2)(3(x-2) - x) \leq 0$$

$$(x-2)(2x-6) \leq 0$$

$$2(x-2)(x-3) \leq 0$$

choose **D**

⑥ $y = 4\sin^{-1}\frac{x}{3}$

$$-1 \leq x \leq 1$$

choose **A**

$$-3 \leq x \leq 3$$

⑦ $R = \sqrt{6^2 + 4^2}$

$$= \sqrt{52}$$

$$= 2\sqrt{13}$$

choose **C**

⑧ $\frac{d^2x}{dt^2} = 10 - 2x^3$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = 10 - 2x^3$$

$$\frac{1}{2}v^2 = 10x - \frac{1}{2}x^4$$

$$v^2 = 20x - x^4 + C$$

at $x = -1$, $v = 0$

$$0 = 20(-1) - (-1)^4 + C$$

$$C = 21$$

$$v^2 = 20x - x^4 + 21$$

choose **B**

⑨ using $\cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}$
 $= 2\cos^2 \frac{\theta}{2} - 1$

$$2\cos^2 \frac{\theta}{2} = \cos \theta + 1$$

$$\cos^2 \frac{\theta}{2} = \frac{1}{2} \cos \theta + \frac{1}{2}$$

so $y = \cos^2 \frac{\theta}{2}$ has amplitude $\frac{1}{2}$, range $0 \leq y \leq 1$,
 period 2π
 choose **B**

⑩ $(1+z+z^2)^5 = (1+z(1+z))^5$

$$\text{now } (1+z(1+z))^5 = {}^5C_0 + {}^5C_1 z(1+z) + {}^5C_2 z^2(1+z)^2 + {}^5C_3 z^3(1+z)^3 \\ + {}^5C_4 z^4(1+z)^4 + {}^5C_5 z^5(1+z)^5$$

terms in z^3 come from ${}^5C_2 z^2(1+z)^2$ and ${}^5C_3 z^3(1+z)$

$$\text{term in } z^3 = ({}^5C_2 \times 2 + {}^5C_3 \times 1) z^3$$

$$\text{coefficient} = 10 \times 2 + 10 \times 1 \\ = 30$$

choose **C**

Section II

⑪ (a) $\sin \frac{\pi}{8} \cos \frac{\pi}{8} = \frac{1}{2} \sin \frac{\pi}{4}$

$$= \frac{1}{2\sqrt{2}}$$

$$= \frac{\sqrt{2}}{4}$$

(b) $\sin^{-1}(\sin \frac{4\pi}{3}) = \sin^{-1}(-\frac{\sqrt{3}}{2})$
 $= -\frac{\pi}{3}$

(c) $\lim_{x \rightarrow 0} \frac{\tan \frac{x}{2}}{x} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{\tan \frac{x}{2}}{\frac{x}{2}}$
 $= \frac{1}{2}$

(d) (i) $\int \frac{4x}{16+x^2} dx = 2 \int \frac{2x}{16+x^2} dx$
 $= 2 \ln(16+x^2) + C$

(ii) $\int \frac{3}{9+x^2} dx = 3 \times \frac{1}{3} \tan^{-1} \frac{x}{3}$
 $= \tan^{-1} \frac{x}{3} + C$

(iii) $\int \frac{-1}{\sqrt{25+x}} dx = -1 \int (25+x)^{-\frac{1}{2}} dx$
 $= -1 \times \frac{(25+x)^{\frac{1}{2}}}{\frac{1}{2}}$
 $= -2\sqrt{25+x} + C$

(e) $\theta = n\pi + (-1)^n \sin^{-1}(-\frac{1}{2})$
 $= n\pi + (-1)^n (-\frac{\pi}{6})$

(f) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{bc + ac + ab}{abc}$
 $= \frac{-\frac{5}{3}}{-(-\frac{5}{8})}$
 $= -\frac{5}{8}$

$$(9) (4x+3)^4 = (4x)^4 + {}^4C_1 (4x)^3(3) + {}^4C_2 (4x)^2(3)^2 + {}^4C_3 (4x)(3)^3 + 3^4$$

$$= 256x^4 + 768x^3 + 864x^2 + 432x + 81$$

the greatest coefficient is 864

(h) $\angle TAB = \angle ACB$ (angle in the alternate segment)
 $\angle TAB = \angle ABC$ (alternate angles, $AT \parallel CB$)
 $AC = AB$ (sides opposite equal angles)

15

$$(12) (a) (i) T = Ae^{-kt} - 8$$

$$\frac{dT}{dt} = -kAe^{-kt}$$

$$= -k(T+8)$$

$$(ii) \text{ at } t=0, T=40$$

$$40 = A - 8$$

$$A = 48$$

$$\text{at } t=30, T=30$$

$$30 = 48e^{-30k} - 8$$

$$\frac{38}{48} = e^{-30k}$$

$$-30k = \log_e\left(\frac{19}{24}\right)$$

$$k = -\frac{1}{30} \log_e\left(\frac{19}{24}\right)$$

$$= \frac{1}{30} \log_e\left(\frac{24}{19}\right)$$

$$(iii) \text{ if } T=0$$

$$0 = 48e^{-kt} - 8$$

$$\frac{8}{48} = e^{-kt}$$

$$-kt = \log_e\left(\frac{1}{6}\right)$$

$$t = -\frac{1}{k} \log_e\left(\frac{1}{6}\right)$$

$$= 230.091... \text{ minutes}$$

$$= 3.83... \text{ h}$$

$$\approx 4 \text{ h}$$

$$(iv) \text{ as } t \rightarrow \infty$$

$$T \rightarrow -8$$

$$(b) (i) \frac{1}{2} (x - \sin x) = \frac{1}{2} \times \frac{1}{2} \pi \quad \checkmark$$

$$x - \sin x = \frac{\pi}{2}$$

$$\sin x = x - \frac{\pi}{2}$$

$$(ii) \text{ let } f(x) = \sin x - x + \frac{\pi}{2}$$

$$f'(x) = \cos x - 1 \quad \checkmark$$

$$2.1 = 2 - \frac{\sin 2 - 2 + \frac{\pi}{2}}{\cos 2 - 1}$$

$$= 2.339...$$

$$\approx 2.34 \quad \checkmark$$

$$(c) (i) \int_0^1 \frac{x}{(3x+1)^2} dx$$

$$= \int_0^1 \frac{\frac{u-1}{3}}{u^2} \times \frac{du}{3}$$

$$= \frac{1}{9} \int_0^1 \frac{u-1}{u^2} du$$

$$= \frac{1}{9} \int_0^1 \frac{1}{u} - u^{-2} du$$

$$= \frac{1}{9} \left[\ln u + \frac{1}{u} \right]_0^1$$

$$= \frac{1}{9} \left(\ln 4 + \frac{1}{4} - (\ln 1 + 1) \right)$$

$$= \frac{1}{9} \left(2 \ln 2 - \frac{3}{4} \right)$$

$$= \frac{2}{9} \ln 2 - \frac{1}{12}$$

$$\text{let } u = 3x+1$$

$$\frac{du}{dx} = 3$$

$$du = 3 dx$$

$$3x = u - 1$$

$$x = \frac{u-1}{3}$$

$$\text{if } x = 0$$

$$u = 1$$

$$\text{if } x = 1$$

$$u = 4$$

$$(ii) V = \pi \int_0^1 \left(\frac{6\sqrt{x}}{3x+1} \right)^2 dx$$

$$= \pi \int_0^1 \frac{36x}{(3x+1)^2} dx$$

$$= 36\pi \times \left(\frac{2}{9} \ln 2 - \frac{1}{12} \right)$$

$$= \pi (8 \ln 2 - 3)$$

cubic
units

$$(d) (i) \tan \theta = \frac{h}{50}$$

$$h = 50 \tan \theta$$

$$\frac{dh}{d\theta} = \frac{50 \sec^2 \theta}{\cos^2 \theta} \quad \checkmark$$

$$(ii) \frac{d\theta}{dt} = \frac{d\theta}{dh} \times \frac{dh}{dt}$$

$$= \frac{\cos^2 \theta}{50} \times \frac{125}{3}$$

$$= \frac{1}{50} \times \frac{125}{3}$$

$$= \frac{5}{3462} \text{ radians/s}$$

$$150 \text{ km/h}$$

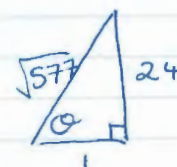
$$= \frac{150 \times 1000}{60 \times 60}$$

$$= \frac{125}{3} \text{ m/s}$$

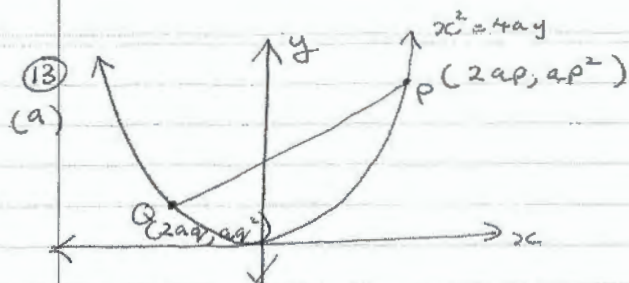
$$\text{if } h = 1200$$

$$\tan \theta = \frac{1200}{50}$$

$$= 24$$



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$$(i) M = \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right)$$

$$= \left(a(p+q), a \frac{(p^2 + q^2)}{2} \right)$$

$$(ii) m = \frac{ap^2 - aq^2}{2ap - 2aq}$$

$$= \frac{p^2 - q^2}{2(p - q)}$$

$$= \frac{(p - q)(p + q)}{2(p - q)}$$

$$= \frac{p + q}{2}, p + q$$

equation is $y - ap^2 = \frac{1}{2}(p+q)(x - 2ap)$

$$y = \frac{1}{2}(p+q)x - apq$$

(iii) if the chord passes through $(0, 2a)$

$$2a = -apq$$

$$pq = -2$$

so $x^2 = a^2(p+q)^2$

$$= a^2(p^2 + q^2 + 2pq)$$

$$= a^2 \left(\frac{2y}{a} - 4 \right)$$

$$= 2a(y - 2a)$$

(b) (i) $x = 2 + \cos^2 t$

$$\dot{x} = 2 \cos t \times -\sin t$$

$$= -2 \sin t \cos t$$

$$= -\sin 2t$$

$$\ddot{x} = -2 \cos 2t$$

$$= -2(2\cos^2 t - 1)$$

$$= -4\cos^2 t + 2$$

$$= -4(x - 2) + 2$$

$$= 10 - 4x \text{ as required}$$

OR

$$x = 2 + \cos^2 t$$

$$= 2 + \frac{1}{2}(\cos 2t + 1)$$

$$= 2\frac{1}{2} + \frac{1}{2}\cos 2t$$

$$\dot{x} = -\sin 2t$$

$$\ddot{x} = -2\cos 2t$$

$$= -2(x - 2\frac{1}{2}) \times 2$$

$$= 10 - 4x \text{ as required}$$

(ii) $\ddot{x} = 10 - 4x$

$$= -4(x - 2\frac{1}{2})$$

which is of the form $\ddot{x} = -n^2(x - x_0)$
(acceleration is proportional to displacement but in the opposite direction)

OR

$$x = 2\frac{1}{2} + \frac{1}{2}\cos 2t$$

which is just a transformation of $x = \cos t$
so is simple harmonic

(iii) centre: $x = 2\frac{1}{2}$ period = $\frac{2\pi}{2}$
 $= \pi$

amplitude = $\frac{1}{2}$

✓ one correct
✓ three correct

(c) $P(x) = x^3 - mx^2 + mx - 1$

(i) $P(1) = 1 - m + m - 1 = 0$
 so $(x-1)$ is a factor

(ii)
$$\begin{array}{r} x^2 + (1-m)x + 1 \\ x-1 \overline{) x^3 - mx^2 + mx - 1} \\ \underline{x^3 - x^2} \\ (1-m)x^2 + mx - 1 \\ \underline{(1-m)x^2 - x + m} \\ x - 1 \\ \underline{x - 1} \\ 0 \end{array}$$

a solution by inspection is fine

the quadratic factor is $x^2 + (1-m)x + 1$

(iii) for real roots we need

$$\begin{aligned} (1-m)^2 - 4(1)(1) &\geq 0 \\ m^2 - 2m + 1 - 4 &\geq 0 \\ m^2 - 2m - 3 &\geq 0 \\ (m-3)(m+1) &\geq 0 \end{aligned}$$

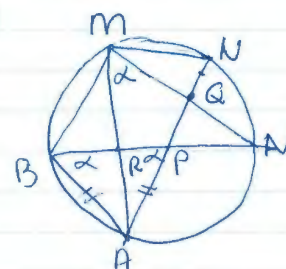


$m \leq -1 \text{ or } m \geq 3$

(iv) If $m = 3$, $P(x) = x^3 - 3x^2 + 3x - 1 = (x-1)^3$

This is the graph of $y = x^3$ shifted 1 unit to the right.

(14) (a)



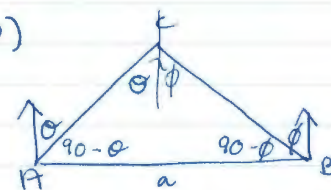
let $\angle ABP = x$

then $\angle PPB = \alpha$ (angles opposite equal sides)

$\angle AMQ = \alpha$ (angles at the circumference on arc AN)

Q, P, R, M is a cyclic quadrilateral (exterior angle equals opposite interior angles)

(b)



$\angle ACB = \theta + \phi$

(i) in $\triangle ACB$, $\frac{\sin(\theta + \phi)}{a} = \frac{\sin(90 - \phi)}{AC}$

in $\triangle APC$, $\tan \alpha = \frac{h}{AC}$
 $AC = \frac{h}{\tan \alpha}$

so $\frac{\sin(\theta + \phi)}{a} = \frac{\sin(90 - \phi)}{\frac{h}{\tan \alpha}}$

$$\checkmark h \sin(\theta + \phi) = a \sin(90 - \phi) \tan \alpha$$

$$h \sin(\theta + \phi) = a \cos \phi \tan \alpha$$

as required

(ii) from $\triangle APC$ from $\triangle BPC$

$$\cot \alpha = \frac{AC}{h} \quad \cot \beta = \frac{BC}{h}$$

$$AC^2 = h^2 \cot^2 \alpha \quad BC^2 = h^2 \cot^2 \beta$$

in $\triangle ABC$

$$\cos(90 - \theta) = \frac{a^2 + h^2 \cot^2 \alpha - h^2 \cot^2 \beta}{2 \times a \times h \cot \alpha}$$

$$\checkmark \sin \theta = \frac{h^2 (\cot^2 \alpha - \cot^2 \beta) + a^2}{2ha \cot \alpha}$$

$$2ha \cot \alpha \sin \theta = h^2 (\cot^2 \alpha - \cot^2 \beta) + a^2$$

$$h^2 (\cot^2 \alpha - \cot^2 \beta) - 2ha \cot \alpha \sin \theta + a^2 = 0$$

(c) Step 1: let $n = 1$

$$f(x) = \frac{1}{x}$$

$$f'(x) = -\frac{1}{x^2}$$

$$\text{now } f^{(1)}(x) = \frac{(-1)^1 1!}{x^{1+1}} = -\frac{1}{x^2} \quad \text{as required}$$

the result is true for $n = 1$

Step 2: suppose n is a positive integer for which the result is true

$$\text{that is } f^{(k)}(x) = \frac{(-1)^k k!}{x^{k+1}} = (-1)^k k! x^{-(k+1)}$$

we now prove the result is true for $n = k+1$, that is we prove that

$$f^{(k+1)}(x) = \frac{(-1)^{k+1} (k+1)!}{x^{k+2}}$$

$$\begin{aligned} \text{now } f^{(k+1)}(x) &= -(k+1) (-1)^k k! x^{-(k+1)-1} \quad \text{by } * \\ &= (-1)^{k+1} (k+1)! x^{-(k+2)} \\ &= \frac{(-1)^{k+1} (k+1)!}{x^{k+2}} \quad \text{as required} \end{aligned}$$

so by the principle of mathematical induction the result is true for all positive integers n .

(14) (d) $x = vt \cos \theta$
 $y = vt \sin \theta - \frac{1}{2} g t^2$

(i) at $x = a$, $y = h$
 so $a = vt \cos \theta$
 $t = \frac{a}{v \cos \theta}$

and $h = \cancel{v} \times \frac{a}{\cancel{v} \cos \theta} \sin \theta - \frac{1}{2} g \left(\frac{a}{v \cos \theta} \right)^2$ ✓

$$h = a \tan \theta - \frac{\frac{1}{2} g a^2}{v^2 \cos^2 \theta}$$

$$\frac{g a^2 \sec^2 \theta}{2 v^2} = a \tan \theta - h$$

$$2 v^2 = \frac{g a^2 \sec^2 \theta}{a \tan \theta - h}$$

$$v^2 = \frac{g a^2 \sec^2 \theta}{2 (a \tan \theta - h)} = \frac{g a^2 (1 + \tan^2 \theta)}{2 (a \tan \theta - h)} \quad (i)$$

(ii) Similarly, $v^2 = \frac{g b^2 (1 + \tan^2 \theta)}{2 (b \tan \theta - h)} \quad (ii)$

equating (i) + (ii)

$$\frac{g b^2 (1 + \tan^2 \theta)}{2 (b \tan \theta - h)} = \frac{g a^2 (1 + \tan^2 \theta)}{2 (a \tan \theta - h)}$$

$$b^2 (a \tan \theta - h) = a^2 (b \tan \theta - h)$$

$$a b^2 \tan \theta - b^2 h = a^2 b \tan \theta - a^2 h$$

$$a b^2 \tan \theta - a^2 b \tan \theta = b^2 h - a^2 h$$

$$a b \tan \theta (b - a) = h (b^2 - a^2)$$

$$a b \tan \theta = \frac{h (b - a) (b + a)}{b - a}$$

$$\tan \theta = \frac{h (b + a)}{a b}$$

(iii) $\tan \theta = \frac{h b}{a b} + \frac{h a}{a b}$

$$= \frac{h}{a} + \frac{h}{b}$$

$$= \tan \alpha + \tan \beta \quad \text{as required}$$

